Aryaman Sharma

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Education

Carnegie Mellon University — Pittsburgh, PA
Master of Computational Data Science (MCDS); GPA: 4.33/4.33 — *Aug. 2026 – Dec. 2027*

- Coursework: Machine Learning, Deep Learning Systems (10-714), Generative AI (10-623)

Netaji Subhas University of Technology — Delhi, India
Bachelor of Technology, Computer Engineering; CGPA: 8.70/10 — *May 2023*

Industry Experience

Fast Retailing (UNIQLO) — Tokyo, Japan
Machine Learning Engineer — *Sep. 2023 – Jul. 2026*

- Built and shipped time-series demand-forecasting models covering 4M+ SKUs across 40 business units, engineering temporal and categorical features that contributed to a USD 170M+ (JPY 26B) annual reduction in stockouts.
- Adapted the forecaster to new-product cold start by extracting product attributes (category, sleeve length, pattern) from product images and descriptions using multimodal LLMs.
- Fine-tuned GPT-4.1 (SFT, Azure OpenAI) for Japanese-to-English product copy translation, building the training-data preparation pipeline and runtime guardrails (prohibited terms, term mappings), cutting translation costs by USD 130K+ (JPY 20M) per year.
- Built a supply-chain production-adjustment assistant using the OpenAI Agents SDK (GCP, Microsoft Teams), automating decisions over 600MB+ of relational data and cutting decision cycles from 3 weeks to 2 days.
- Built CockpitAI, an interactive multi-agent system for querying business reports (parallel analysis agents, hybrid SQL + pgvector retrieval, GPT-5 vision Q&A), cutting multimodal retrieval latency by 60%.

Research Experience

University of South Carolina, AI Institute (AIISC) — Remote
Research Associate (Collaborator), Prof. Amit Sheth — *Mar. 2025 – Dec. 2025*

- Co-developed **CausalTrace** (AAAI 2026, IAAI), a neurosymbolic causal-analysis agent for a manufacturing copilot, combining causal discovery, counterfactual reasoning, and root-cause analysis over knowledge graphs.
- Developed the evaluation protocol: an LLM-as-judge harness (GPT-4o-mini, LLaMA3-70B) validated against 6 human domain experts, with inter-rater agreement of Cohen's Kappa 0.91 and retrieval scores of MAP@3 94%, PR@2 97%, MRR 0.92, ROUGE-1 0.91.
- Built ForeSight, an hourly production forecasting model (35% lower MSE than an LSTM baseline), and InfoGuide, a conversational Q&A agent over causal graphs with hybrid neural-symbolic retrieval.

Vanderbilt University, SCOPE Lab — Nashville, TN
Summer Research Intern (NSF-funded), Prof. Abhishek Dubey — *Jun. 2022 – Aug. 2022*

- Reached 96% accuracy on stop-level transit occupancy prediction with Transformer and LSTM models trained on spatiotemporal ridership data, used to inform resource allocation for paratransit services.

Publications

Google Scholar

CausalTrace: A Neurosymbolic Causal Analysis Agent for Smart Manufacturing. AAAI 2026, IAAI Track.

Designing Equitable Transit Networks. ACM EAAMO 2023 (non-archival track).

Technical Skills

Programming: Python, C++, SQL, R, JavaScript (React)
ML Libraries: PyTorch, TensorFlow, scikit-learn (sklearn), Hugging Face Transformers, NumPy, FAISS
ML Methods: Time Series Forecasting, Causal Inference, Multimodal LLMs, LLM Fine-Tuning (SFT), RLHF/DPO, RAG, Multi-Agent Systems (LangGraph, AutoGen, OpenAI Agents SDK), MCP, DSPy
Evaluation: Experiment Design, Human Evaluation, LLM-as-Judge, Inter-Rater Agreement (Cohen's Kappa), Ranking Metrics (MAP, MRR)
Systems & Cloud: GCP (Vertex AI), Azure OpenAI, PostgreSQL/pgvector, FastAPI, Docker, Git, Linux, pytest
Languages: English (Fluent), Hindi (Native), French (DELF B2), Japanese (JLPT N2)